1-8 | 9 | 10 | 11 | 12 | 13 | 14 | 15
20 | 8 | 10 | 0 | 11 | 6 | 0 | 0

Econometrics (25117): Midterm 1, 2026–2027
Duration: 180 minutes
Monday, 9 November 2026

Name and surname: Gloria Carrera UPF ID: U273909

Answer all questions. Questions 1–8 have one correct option each: 2.5 points for a correct answer, 0 for a wrong or blank answer. Questions 9–15 are worth 80 points; show calculations and briefly explain your reasoning on separate answer paper. The total is 100 points; the grade out of 10 is the total divided by 10. For inference, use the large-sample standard normal approximation. The two-sided critical values are 1.96 at 5% and 2.576 at 1%. Exact fractions, radicals or sensible rounding are acceptable. All numerical examples are hypothetical.

Multiple-choice questions

1. Let $Y_1, \ldots, Y_n$ be i.i.d. with mean μ and variance $0 < \sigma^2 < \infty$. Consider $\widetilde{\mu}_n = Y_1$, which uses only the first observation, regardless of n. This estimator is:

   (a) unbiased and consistent.
   (b) unbiased and inconsistent.
   (c) biased and consistent.
   (d) biased and inconsistent.

2. Which statement always holds when the required moments exist?

   (a) Zero covariance implies independence.
   (b) Zero covariance implies $\mathrm{E}[Y \mid X] = \mathrm{E}[Y]$.
   (c) Independence implies zero covariance.
   (d) Perfect positive correlation implies independence.

3. For i.i.d. observations with mean μ and variance $0 < \sigma^2 < \infty$, the central limit theorem states that, as $n \to \infty$:

   (a) $\bar{Y}$ has exactly a normal distribution for every n.
   (b) $\sqrt{n}(\bar{Y} - \mu)/\sigma \xrightarrow{d} N(0,1)$.
   (c) $(\bar{Y} - \mu)/\sigma \xrightarrow{d} N(0,1)$.
   (d) $\bar{Y} \xrightarrow{p} 0$, whatever the value of μ.

4. A Type I error occurs when a test:

   (a) rejects a true null hypothesis.
   (b) does not reject a false null hypothesis.
   (c) rejects a false null hypothesis.
   (d) does not reject a true null hypothesis.

5. A large-sample two-sided test uses an estimate $\bar{y} = 3$ and $SE(\bar{Y}) = 1$. The null $H_0 : \mu_Y = 0$ is rejected at 5%. For any hypothesized value μ_0, reject when $|t| > 1.96$. What is the smallest μ_0 for which $H_0 : \mu_Y = \mu_0$ would not be rejected?

   (a) 0.
   (b) 1.96.
   (c) 4.96.
   (d) 1.04.

6. In an OLS regression with an intercept and a nonconstant regressor, which statement is necessarily true, apart from numerical rounding?

   (a) $\sum_i \widehat{U}_i = 0$ proves that $\mathrm{E}[U_i \mid X_i] = 0$.
   (b) $\sum_i \widehat{U}_i = 0$ and $\sum_i X_i \widehat{U}_i = 0$, even if the regressor is endogenous.
   (c) $U_i = \widehat{U}_i$ for every observation.
   (d) Every individual residual equals zero.

7. For i.i.d. observations with a finite, positive population variance, the standard deviation of the sample mean is $\sigma_Y/\sqrt{n}$. If the sample size increases from 100 to 400 while the population distribution stays the same, the standard deviation of the sample mean:

   (a) is divided by four.
   (b) is multiplied by two.
   (c) is unchanged.
   (d) is divided by two.

8. For simple OLS with an intercept, $n > 2$ and $SSR > 0$, which statement correctly distinguishes the standard error of the regression (SER) from the root mean squared residual ($RMSE$)?

   (a) $SER = \sqrt{SSR/n}$; $RMSE = \sqrt{SSR/(n-2)}$.
   (b) $SER = RMSE$ for every finite sample.
   (c) $SER = \sqrt{SSR/(n-2)}$; $RMSE = \sqrt{SSR/n}$.
   (d) Both equal $\sqrt{SSR/(n-1)}$.

1–8 | 9 | 10 | 11 | 12 | 13 | 14 | 15
20 | 7 | 9 | 24 | 7 | 6.5 | 0 | 0

im not sure about the UPF id sorry!

Econometrics (25117): Midterm 1, 2026–2027

Duration: 180 minutes

Monday, 9 November 2026

DNI = 47598020A

Name and surname: Natalia Jacek UPF ID: U223924

Answer all questions. Questions 1–8 have one correct option each: 2.5 points for a correct answer, 0 for a wrong or blank answer. Questions 9–15 are worth 80 points; show calculations and briefly explain your reasoning on separate answer paper. The total is 100 points; the grade out of 10 is the total divided by 10. For inference, use the large-sample standard normal approximation. The two-sided critical values are 1.96 at 5% and 2.576 at 1%. Exact fractions, radicals or sensible rounding are acceptable. All numerical examples are hypothetical.

consistent → $\bar{Y} \to \mu_N$ as $n \to \infty$

unbiased → $E(Y_i)$ $\quad Y_1 = \tilde{\mu}_N$

Multiple-choice questions

1. Let $Y_1, \ldots, Y_n$ be i.i.d. with mean μ and variance $0 < \sigma^2 < \infty$. Consider $\tilde{\mu}_n = Y_1$, which uses only the first observation, regardless of n. This estimator is:

   (a) unbiased and consistent. (b) unbiased and inconsistent.
   (c) biased and consistent. (d) biased and inconsistent.

2. Which statement always holds when the required moments exist?

   (a) Zero covariance implies independence. (b) Zero covariance implies $E[Y \mid X] = E[Y]$.
   (c) Independence implies zero covariance. (d) Perfect positive correlation implies independence.

   independence ⇏ zero cov

3. For i.i.d. observations with mean μ and variance $0 < \sigma^2 < \infty$, the central limit theorem states that, as $n \to \infty$:

   (a) $\bar{Y}$ has exactly a normal distribution for every n. (b) $\sqrt{n}(\bar{Y} - \mu)/\sigma \xrightarrow{d} N(0,1)$.
   (c) $(\bar{Y} - \mu)/\sigma \xrightarrow{d} N(0,1)$. (d) $\bar{Y} \xrightarrow{p} 0$, whatever the value of μ.

   $\frac{\bar{y}-\mu}{SD} = \frac{\bar{y}-\mu}{\sigma/\sqrt{n}} \to N(0,1)$

4. A Type I error occurs when a test:

   (a) rejects a true null hypothesis. (b) does not reject a false null hypothesis.
   (c) rejects a false null hypothesis. (d) does not reject a true null hypothesis.

   false (+) we accept/reject type I error; false (+)

   $\frac{(\bar{y}-\mu)\sqrt{n}}{\sigma}$

5. A large-sample two-sided test uses an estimate $\bar{y} = 3$ and $SE(\bar{Y}) = 1$. The null $H_0 : \mu_Y = 0$ is rejected at 5%. For any hypothesized value μ_0, reject when $|t| > 1.96$. What is the smallest μ_0 for which $H_0 : \mu_Y = \mu_0$ would not be rejected?

   $-1.96 < \frac{(\bar{y}-\mu)\sqrt{n}}{\sigma} < 1.96$ → $-1.96 < \frac{(\bar{y}-\mu)}{1} < 1.96$ → $-1.96 - 3 < -\mu < 1.96 - 3$

   $-4.96 < -\mu < -1.04$

   $4.96 < \mu < 1.04$ reject if $\mu < 1.04$

   (a) 0. (b) 1.96.
   (c) 4.96. (d) 1.04.

6. In an OLS regression with an intercept and a nonconstant regressor, which statement is necessarily true, apart from numerical rounding?

   $E(u) = 0 \to E(E(u_i \mid x_i)) = 0$

   (a) $\sum_i \hat{U}_i = 0$ proves that $E[U_i \mid X_i] = 0$. (b) $\sum_i \hat{U}_i = 0$ and $\sum_i X_i \hat{U}_i = 0$, even if the regressor is endogenous.
   (c) $U_i = \hat{U}_i$ for every observation. (d) Every individual residual equals zero.

7. For i.i.d. observations with a finite, positive population variance, the standard deviation of the sample mean is $\sigma_Y/\sqrt{n}$. If the sample size increases from 100 to 400 while the population distribution stays the same, the standard deviation of the sample mean:

   $\frac{\sigma_Y}{\sqrt{n}} = \frac{\sigma_Y}{\sqrt{100}} \to \frac{\sigma_Y}{\sqrt{400}} = \frac{\sqrt{100}}{\sqrt{400}} = 1/2$

   (a) is divided by four. (b) is multiplied by two.
   (c) is unchanged. (d) is divided by two.

8. For simple OLS with an intercept, $n > 2$ and $SSR > 0$, which statement correctly distinguishes the standard error of the regression (SER) from the root mean squared residual ($RMSE$)?

   (a) $SER = \sqrt{SSR/n}$; $RMSE = \sqrt{SSR/(n-2)}$. (b) $SER = RMSE$ for every finite sample.
   (c) $SER = \sqrt{SSR/(n-2)}$; $RMSE = \sqrt{SSR/n}$. (d) Both equal $\sqrt{SSR/(n-1)}$.

   $SER = \sqrt{\frac{SSR}{n-2}}$ $\quad RMSE = \sqrt{\frac{SSR}{n}}$

1-8 | 9 | 10 | 11 | 12 | 13 | 14 | 15
15 | 7 | 8 | 0 | 9 | 7 | 5.5 | 8

Econometrics (25117): Midterm 1, 2026–2027

Duration: 180 minutes

Monday, 9 November 2026

Name and surname: Alex Calls Miras UPF ID: U 2 7 3 9 3 7

Answer all questions. Questions 1–8 have one correct option each: 2.5 points for a correct answer, 0 for a wrong or blank answer. Questions 9–15 are worth 80 points; show calculations and briefly explain your reasoning on separate answer paper. The total is 100 points; the grade out of 10 is the total divided by 10. For inference, use the large-sample standard normal approximation. The two-sided critical values are 1.96 at 5% and 2.576 at 1%. Exact fractions, radicals or sensible rounding are acceptable. All numerical examples are hypothetical.

Multiple-choice questions

1. Let $Y_1, \ldots, Y_n$ be i.i.d. with mean μ and variance $0 < \sigma^2 < \infty$. Consider $\widetilde{\mu}_n = Y_1$, which uses only the first observation, regardless of n. This estimator is:

   (a) unbiased and consistent. (b) unbiased and inconsistent.
   (c) biased and consistent. (d) biased and inconsistent.

2. Which statement always holds when the required moments exist?

   (a) Zero covariance implies independence. (b) Zero covariance implies $E[Y \mid X] = E[Y]$.
   (c) Independence implies zero covariance. (d) Perfect positive correlation implies independence.

3. For i.i.d. observations with mean μ and variance $0 < \sigma^2 < \infty$, the central limit theorem states that, as $n \to \infty$:

   (a) $\bar{Y}$ has exactly a normal distribution for every n. (b) $\sqrt{n}(\bar{Y} - \mu)/\sigma \xrightarrow{d} N(0,1)$.
   (c) $(\bar{Y} - \mu)/\sigma \xrightarrow{d} N(0,1)$. (d) $\bar{Y} \xrightarrow{p} 0$, whatever the value of μ.

4. A Type I error occurs when a test:

   (a) rejects a true null hypothesis. (b) does not reject a false null hypothesis.
   (c) rejects a false null hypothesis. (d) does not reject a true null hypothesis.

5. A large-sample two-sided test uses an estimate $\bar{y} = 3$ and $SE(\bar{Y}) = 1$. The null $H_0 : \mu_Y = 0$ is rejected at 5%. For any hypothesized value μ_0, reject when $|t| > 1.96$. What is the smallest μ_0 for which $H_0 : \mu_Y = \mu_0$ would not be rejected?

   (a) 0. (b) 1.96.
   (c) 4.96. (d) 1.04.

6. In an OLS regression with an intercept and a nonconstant regressor, which statement is necessarily true, apart from numerical rounding?

   (a) $\sum_i \widehat{U}_i = 0$ proves that $E[U_i \mid X_i] = 0$. (b) $\sum_i \widehat{U}_i = 0$ and $\sum_i X_i \widehat{U}_i = 0$, even if the regressor is endogenous.
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7. For i.i.d. observations with a finite, positive population variance, the standard deviation of the sample mean is $\sigma_Y/\sqrt{n}$. If the sample size increases from 100 to 400 while the population distribution stays the same, the standard deviation of the sample mean:

   (a) is divided by four. (b) is multiplied by two.
   (c) is unchanged. (d) is divided by two.

8. For simple OLS with an intercept, $n > 2$ and $SSR > 0$, which statement correctly distinguishes the standard error of the regression (SER) from the root mean squared residual ($RMSE$)?

   (a) $SER = \sqrt{SSR/n}$; $RMSE = \sqrt{SSR/(n-2)}$. (b) $SER = RMSE$ for every finite sample.
   (c) $SER = \sqrt{SSR/(n-2)}$; $RMSE = \sqrt{SSR/n}$. (d) Both equal $\sqrt{SSR/(n-1)}$.

1-8 | 9 | 10 | 11 | 12 | 13 | 14 | 15
20 | 8 | 8 | 12 | 2.5 | 12 | 0 | 0

Econometrics (25117): Midterm 1, 2026–2027

Duration: 180 minutes

Monday, 9 November 2026

Name and surname: Gerard Samin Nogués UPF ID: U273829

Answer all questions. Questions 1–8 have one correct option each: 2.5 points for a correct answer, 0 for a wrong or blank answer. Questions 9–15 are worth 80 points; show calculations and briefly explain your reasoning on separate answer paper. The total is 100 points; the grade out of 10 is the total divided by 10. For inference, use the large-sample standard normal approximation. The two-sided critical values are 1.96 at 5% and 2.576 at 1%. Exact fractions, radicals or sensible rounding are acceptable. All numerical examples are hypothetical.

Multiple-choice questions

1. Let $Y_1, \ldots, Y_n$ be i.i.d. with mean μ and variance $0 < \sigma^2 < \infty$. Consider $\widetilde{\mu}_n = Y_1$, which uses only the first observation, regardless of n. This estimator is:

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   (b) unbiased and inconsistent.
   (c) biased and consistent.
   (d) biased and inconsistent.

2. Which statement always holds when the required moments exist?

   (a) Zero covariance implies independence.
   (b) Zero covariance implies $\mathrm{E}[Y \mid X] = \mathrm{E}[Y]$.
   (c) Independence implies zero covariance.
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3. For i.i.d. observations with mean μ and variance $0 < \sigma^2 < \infty$, the central limit theorem states that, as $n \to \infty$:

   (a) $\bar{Y}$ has exactly a normal distribution for every n.
   (b) $\sqrt{n}(\bar{Y} - \mu)/\sigma \xrightarrow{d} N(0,1)$.
   (c) $(\bar{Y} - \mu)/\sigma \xrightarrow{d} N(0,1)$.
   (d) $\bar{Y} \xrightarrow{p} 0$, whatever the value of μ.

4. A Type I error occurs when a test:

   (a) rejects a true null hypothesis.
   (b) does not reject a false null hypothesis.
   (c) rejects a false null hypothesis.
   (d) does not reject a true null hypothesis.

5. A large-sample two-sided test uses an estimate $\bar{y} = 3$ and $SE(\bar{Y}) = 1$. The null $H_0 : \mu_Y = 0$ is rejected at 5%. For any hypothesized value μ_0, reject when $|t| > 1.96$. What is the smallest μ_0 for which $H_0 : \mu_Y = \mu_0$ would not be rejected?

   $t = \frac{\bar{y} - \mu_0}{SE(\bar{Y})} = 1'96 \; ; \; \mu_0 = 1'04$

   (a) 0.
   (b) 1.96.
   (c) 4.96.
   (d) 1.04.

6. In an OLS regression with an intercept and a nonconstant regressor, which statement is necessarily true, apart from numerical rounding?

   (a) $\sum_i \widehat{U}_i = 0$ proves that $\mathrm{E}[U_i \mid X_i] = 0$.
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   (c) is unchanged.
   (d) is divided by two.

8. For simple OLS with an intercept, $n > 2$ and $SSR > 0$, which statement correctly distinguishes the standard error of the regression (SER) from the root mean squared residual ($RMSE$)?

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1-8 | 9 | 10 | 11 | 12 | 13 | 14 | 15
20 | 10 | 9 | 10 |) | 6.5 | 8 | 9

Econometrics (25117): Midterm 1, 2026–2027

Duration: 180 minutes

Monday, 9 November 2026

Name and surname: Fernando Albacar UPF ID: U 2 7 3 9 5 8

Answer all questions. Questions 1–8 have one correct option each: 2.5 points for a correct answer, 0 for a wrong or blank answer. Questions 9–15 are worth 80 points; show calculations and briefly explain your reasoning on separate answer paper. The total is 100 points; the grade out of 10 is the total divided by 10. For inference, use the large-sample standard normal approximation. The two-sided critical values are 1.96 at 5% and 2.576 at 1%. Exact fractions, radicals or sensible rounding are acceptable. All numerical examples are hypothetical.

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   (c) $(\bar{Y} - \mu)/\sigma \xrightarrow{d} N(0,1)$.
   (d) $\bar{Y} \xrightarrow{p} 0$, whatever the value of μ.

4. A Type I error occurs when a test:

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   (c) rejects a false null hypothesis.
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5. A large-sample two-sided test uses an estimate $\bar{y} = 3$ and $SE(\bar{Y}) = 1$. The null $H_0 : \mu_Y = 0$ is rejected at 5%. For any hypothesized value μ_0, reject when $|t| > 1.96$. What is the smallest μ_0 for which $H_0 : \mu_Y = \mu_0$ would not be rejected?

   $t = 1.96 = \frac{3 - \mu_0}{1}$ $\mu_0 = 1.04$

   (a) 0.
   (b) 1.96.
   (c) 4.96.
   (d) 1.04.

6. In an OLS regression with an intercept and a nonconstant regressor, which statement is necessarily true, apart from numerical rounding?

   (a) $\sum_i \widehat{U}_i = 0$ proves that $\mathrm{E}[U_i \mid X_i] = 0$.
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7. For i.i.d. observations with a finite, positive population variance, the standard deviation of the sample mean is $\sigma_Y/\sqrt{n}$. If the sample size increases from 100 to 400 while the population distribution stays the same, the standard deviation of the sample mean:

   $\sigma_Y = 1$

   $\sigma_{\bar{Y}_1} = \frac{\sigma_Y}{\sqrt{n}}$ $\sigma_{\bar{Y}_2} = \frac{\sigma_Y}{2\sqrt{n}}$

   $\frac{1}{2} = 0.1$ $= 0.05$

   (a) is divided by four.
   (b) is multiplied by two.
   (c) is unchanged.
   (d) is divided by two.

8. For simple OLS with an intercept, $n > 2$ and $SSR > 0$, which statement correctly distinguishes the standard error of the regression (SER) from the root mean squared residual ($RMSE$)?

   (a) $SER = \sqrt{SSR/n}$; $RMSE = \sqrt{SSR/(n-2)}$.
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1-8 | 9 | 10 | 11 | 12 | 13 | 14 | 15
17'5 | 7 | 10 10 | 18 | 0 | 4 | 0 | 0

Econometrics (25117): Midterm 1, 2026–2027

Duration: 180 minutes

Monday, 9 November 2026

Name and surname: Cristina Jiménez UPF ID: U273940

Answer all questions. Questions 1–8 have one correct option each: 2.5 points for a correct answer, 0 for a wrong or blank answer. Questions 9–15 are worth 80 points; show calculations and briefly explain your reasoning on separate answer paper. The total is 100 points; the grade out of 10 is the total divided by 10. For inference, use the large-sample standard normal approximation. The two-sided critical values are 1.96 at 5% and 2.576 at 1%. Exact fractions, radicals or sensible rounding are acceptable. All numerical examples are hypothetical.

Multiple-choice questions

1. Let $Y_1, \ldots, Y_n$ be i.i.d. with mean μ and variance $0 < \sigma^2 < \infty$. Consider $\tilde{\mu}_n = Y_1$, which uses only the first observation, regardless of n. This estimator is:

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   (c) biased and consistent.
   (d) biased and inconsistent.

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   (d) Perfect positive correlation implies independence.

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   (b) $\sqrt{n}(\bar{Y} - \mu)/\sigma \xrightarrow{d} N(0,1)$.
   (c) $(\bar{Y} - \mu)/\sigma \xrightarrow{d} N(0,1)$.
   (d) $\bar{Y} \xrightarrow{p} 0$, whatever the value of μ.

4. A Type I error occurs when a test:

   (a) rejects a true null hypothesis.
   (b) does not reject a false null hypothesis.
   (c) rejects a false null hypothesis.
   (d) does not reject a true null hypothesis.

5. A large-sample two-sided test uses an estimate $\bar{y} = 3$ and $SE(\bar{Y}) = 1$. The null $H_0 : \mu_Y = 0$ is rejected at 5%. For any hypothesized value μ_0, reject when $|t| > 1.96$. What is the smallest μ_0 for which $H_0 : \mu_Y = \mu_0$ would not be rejected?

   (a) 0.
   (b) 1.96.
   (c) 4.96.
   (d) 3.04.

6. In an OLS regression with an intercept and a nonconstant regressor, which statement is necessarily true, apart from numerical rounding?

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   (d) Both equal $\sqrt{SSR/(n-1)}$.